



$$Area = \int_0^4 \int_0^{\frac{x^3}{32}} dy dx = \int_0^4 \frac{x^3}{32} dx = \frac{1}{128} x^4 \Big|_0^4 = \frac{256}{128} = \boxed{2 \text{ in}^2 = A}$$

$$I_x = \int_A y^2 dA = \int_0^4 \int_0^{\frac{x^3}{32}} y^2 dy dx = \int_0^4 \frac{1}{3} \left(\frac{x^3}{32} \right)^3 dx = \int_0^4 \frac{1}{3 \cdot 32^3} x^9 dx$$

$$= \frac{1}{30 \cdot 32^3} 4^{10} = \boxed{1.067 \text{ in}^4 = I_x}$$

$$I_y = \int_A x^2 dA = \int_0^4 \int_0^{\frac{x^3}{32}} x^2 dy dx = \int_0^4 \frac{x^5}{32} dx = \frac{1}{6 \cdot 32} x^6 \Big|_0^4 = \boxed{21.33 \text{ in}^4 = I_y}$$

$$I_{xy} = \int_A xy dA = \int_0^4 \int_0^{\frac{x^3}{32}} xy dy dx = \int_0^4 x \left(\frac{1}{2} \right) \left(\frac{x^3}{32} \right)^2 dx = \frac{x^8}{16 \cdot 32^2} \Big|_0^4 = \boxed{4.0 \text{ in}^4 = I_{xy}}$$

$$J_o = I_x + I_y = \boxed{22.40 \text{ in}^4 = J_o}$$

$$r_{ex} = \sqrt{I_x/A} = \sqrt{1.067/2} = \boxed{0.730 \text{ in} = r_{ex}}$$

$$r_{ey} = \sqrt{I_y/A} = \sqrt{21.33/2} = \boxed{3.266 \text{ in} = r_{ey}}$$

$$I_{MAX} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2} \right)^2 + I_{xy}^2}$$

$$= \frac{1.067 + 21.33}{2} \pm \sqrt{\left(\frac{1.067 - 21.33}{2} \right)^2 + 4^2}$$

$$I_{MAX} = 11.2 \pm 10.89$$

$\Rightarrow$

$$\boxed{I_{MAX} = 22.09 \text{ in}^4}$$

$$\boxed{I_{MIN} = 0.3058 \text{ in}^4}$$

$$2\theta_p = \tan^{-1} \frac{-I_{xy}}{\frac{I_x - I_y}{2}} = \tan^{-1} \frac{-4}{\frac{1.067 - 21.33}{2}} = 21.5^\circ = 2\theta_p$$

$$\boxed{\theta_p = 10.8^\circ}$$